Research Recommender System

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Course: CMPE 256 Recommender Systems

I. Objective

Our objective is to help people learn more about a topic via recommending research papers/ articles to their needs. The goal is to recommend papers for a user to cite by selecting topically relevant cross-domain articles via a custom similarity scoring, not through keyword search only. This custom scoring includes a variety of models, such as TF_IDF only, SVD, NeuMF, PageRank, etc, with a final hybrid recommender combining the best individual performers. Our primary goal is to provide cross-domain recommendations based on the users exact needs via citation data, category names and keywords in the abstract. If a user is looking for research regarding stock market modelling, we should recommend machine learning papers related to time-series and forecasting, as well as economic articles related to market modelling. This project proposes a hybrid recommender system that combines content-based filtering with collaborative filtering derived from citation-based interactions. This recommender will be useful for anyone interested in cross-domain articles to do further research and cite for their own academic research purposes.

II. Data Description & EDA

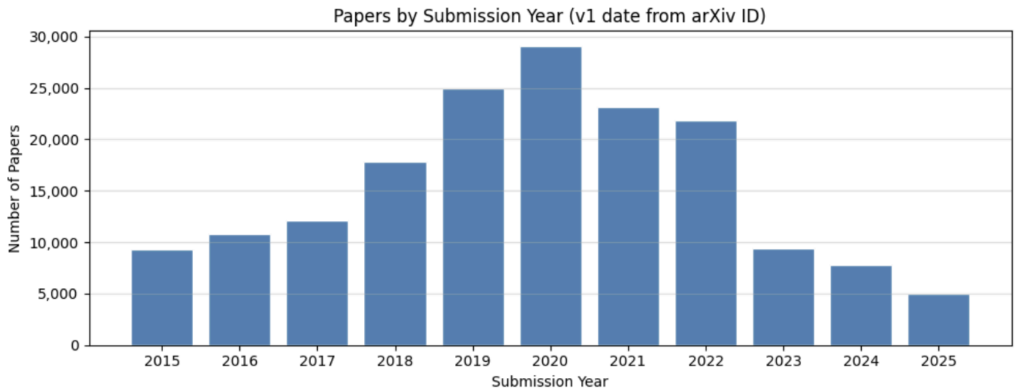
Our dataset, as linked below, is Cornell's arXiv dataset, which is available to the public and meant for educational and research purposes.[1] It provides access to 1.7M scholarly articles, and their information neatly organized into a tabular dataset containing columns such as: titles, abstracts, authors, categories, update_date, etc. The categories include Computer Science, Mathematics, Physics, Electrical Engineering, Statistics, Finance, Biology, Economics, and a few more. The breakdown of the original dataset is as follows: The original arXiv dataset is composed of 51% Physics, 22% Mathematics, 19% Computer Science, 1.8% Statistics, 1.3% Electrical Engineering & Systems Science, 1.1% Quantitative Biology, 0.4% Quantitative Finance, and 0.2% Economics. Unfortunately, this dataset does not include the citation data; The original dataset with these data along with the full text, sections etc is stored under the title unarXive, linked below, but unfortunately requires additional permissions.[2] To bypass this, we were able to extract the citation data via Semantic Scholar API and saved the data into a citation dataframe containing the source id and the citation id.

Due to the time in creating the citation data (edges), we chose to limit our dataset to ~486k rows, with 1M+ citation links. Additionally, as some models cannot handle if a paper cites no other paper (cold start problem), so we added a cold start filter to remove these cases, creating

[1] https://www.kaggle.com/datasets/Cornell-University/arxiv
[2] https://zenodo.org/records/7752754

a 171,289 x 171,289 paper-paper citation matrix, with 951,597 citation links (edges). After applying the cold-start filter, the final distribution was: Computer Science (37.2%) followed by Physics (18.3%), Electrical Engineering (14.1%), Mathematics (13.2%), and Statistics (10.0%), while Biology (3.8%), Finance (2.1%), and Economics (1.5%).

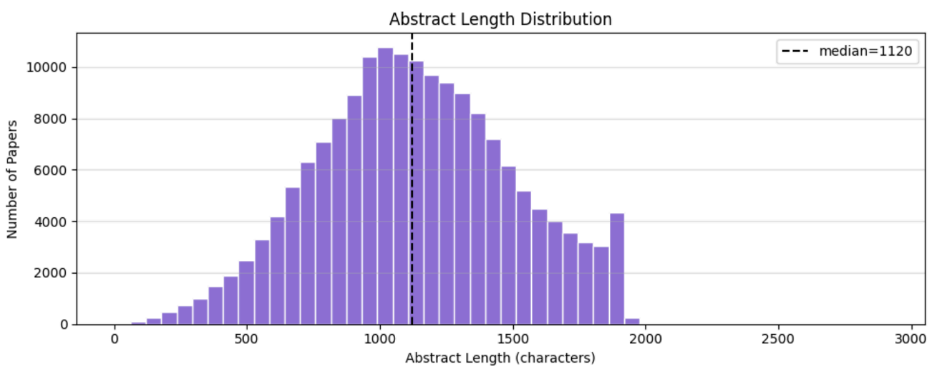
EDA* after Cold Start Filter:



year	
2015	9268
2016	10809
2017	12154
2018	17807
2019	24985
2020	29099
2021	23201
2022	21808
2023	9437
2024	7770
2025	4951

Graph 1

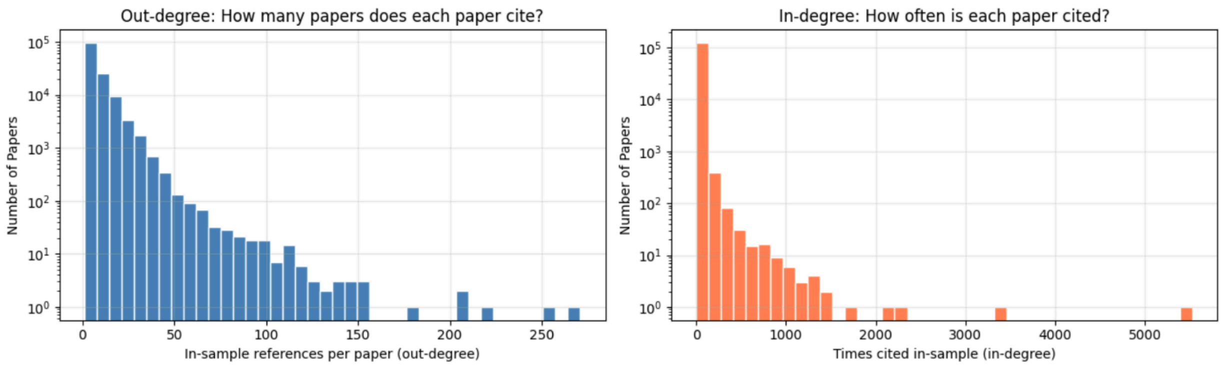
★ **Dataset Submission Year - Graph 1 Analysis**
- ○ The final dataset contains research articles published between 2015 and 2025
- ○ Most articles are in the 2019-2021 timeframe



abstract	length
count	171289
mean	1132
std	371
min	3
25%	874
50%	1120
75%	1386
max	2906

Graph 2

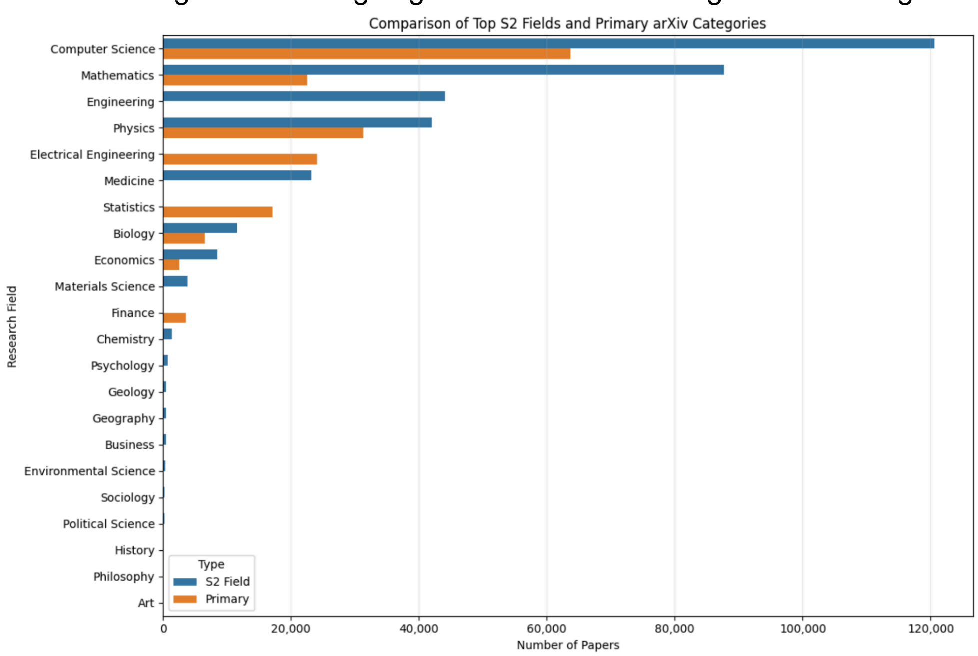
★ Article Characteristics - Graph 2 Analysis
- ○ Papers contained relatively detailed summaries, with a median abstract length of 1,120 words, supporting content-based recommendation methods.



Graph 3

★ **<u>Citation Network Structure - Graph 3 Analysis & General Citation Analysis</u>**

- The citation interaction matrix was extremely sparse (99.9968%), indicating that only a very small fraction of possible paper-to-paper citation relationships exist.
- Approximately 26.5% of papers experienced cold-start conditions, meaning they were not cited by any other paper in the dataset.
- 137,750 papers (80.4%) cited at least one other paper, while 123,989 papers (72.4%) received at least one citation.
- Citation behavior remained relatively connected overall, with papers containing an average of 6.91 outgoing citations and receiving 7.67 incoming citation



Graph 4

★ **<u>Category & Domain Distribution - Graph 4 Analysis</u>**

- Comparison between primary arXiv categories and Semantic Scholar (S2) fields of study showed different patterns of domain representation.

- Primary categories were more concentrated in Computer Science, Physics, Electrical Engineering, and Mathematics, reflecting the original arXiv collection.
- Semantic Scholar fields showed broader interdisciplinary coverage, with relatively stronger representation across Statistics, Electrical Engineering, Finance, and additional domains such as Chemistry, Psychology, Geology, Geography, Business, Political Science, History, and Arts.
- This broader field coverage supports the project goal of enabling cross-domain recommendation and discovery beyond a paper's original research category

III. Methodology

Data Pipeline (creating final dataset):

- ★ arXiv OAI-PMH harvest
  - metadataPrefix="arXiv" for clean <categories> elements.
  - 21 parallel workers (3 sets x 7 years). Atomic checkpoint writes per (set, year).
- ★ Semantic Scholar citation fetch
  - 500 IDs per batch, shared token-bucket rate limiter, incremental checkpointing to Google Drive.
  - Staleness check prevents using a checkpoint from a different OAI-PMH run.
- ★ In-sample filtering
  - Keep edges where both source and target paper are in the collected dataset. astype(str) coercion prevents float-parse bug on IDs like "2301.12345".
- ★ Cold-start filter
  - drop papers with total degree < 2.
- ★ Integer index
  - 0-based paper_idx for direct scipy/ALS/NeuMF compatibility.
- ★ Feature engineering:
  - text = title + title + abstract (title doubled for higher TF-IDF weight)
  - cite_pop = normalized in-sample citation count (0–1)
  - s2_fields = Semantic Scholar fields of stud

Models:

Model	**Signal**
SVD (Baseline)	Matrix factorization of the paper-paper citation matrix to learn latent citation relationships
TF-IDF	Computes cosine similarity using TF-IDF features extracted from paper title and abstract text
NeuMF	Learns nonlinear latent citation patterns using neural embeddings and interaction layers
PageRank	Computes global paper importance through recursive citation propagation across the citation graph

Monte PPR	Estimates locally relevant papers through random walks starting from a query paper
GMF	Learns non-linear paper embeddings using neural matrix factorization on citation interactions
Bandits	Balances exploitation of strong recommendations with exploration of potentially relevant unseen papers
Hybrid 1	Weighted combination of TF-IDF + SVD
Hybrid 2	Weighted combination of TF-IDF + Monte PPR

Overall, we applied the following types of recommender systems: Content-based, Collaborative Filtering, Graph Network (global & local) and two Hybrid models.

Model Breakdown:

Our goal is to try many algorithms and create one hybrid recommender to combine all sorts of signals from our algorithms. Since we do not have any sort of global mean/ popularity metric, we used SVD as our baseline model. To implement **SVD**, we constructed a sparse paper–paper citation matrix from the citation edge list, where each entry is 1 if one paper cites another. We then applied truncated matrix factorization to decompose this interaction matrix into low-dimensional latent embeddings for each paper. This is appropriate for the problem because citation behavior encodes implicit relationships between papers and SVD captures hidden structural patterns such as shared research communities or methodological similarity even when papers do not directly cite each other.

Following SVD, we tested out a content based approach, **TF-IDF**. We used the text & abstract text and converted the text into vectors and applied TF-IDF to discover what words are most important in each article. After this, we used cosine similarity to determine which articles were most similar to each other. After this testing, we moved towards collaborative filtering, covering a range of advanced methods to a valuable cross-domain recommender system.

Next, we focused on additional collaborative filtering models such as **GMF, Epsilon-Greedy, and NeuMF**. We implemented **Generalized Matrix Factorization (GMF)**, a neural recommendation model designed to learn latent interactions between papers through embedding representations. Unlike traditional SVD, which relies on linear matrix decomposition, GMF learns embeddings directly through a neural framework, allowing the model to better capture nonlinear citation relationships between papers. Each paper was represented using a learned embedding vector, and recommendation relevance was computed using the element-wise product between source and target paper embeddings followed by a prediction layer. The model was trained using implicit citation feedback with binary cross entropy loss and negative sampling, where randomly selected non cited papers acted as negative examples. This helped to enable GMF to learn dense latent representations that captured topical overlap, methodological similarity, and research community structure all the while remaining computationally efficient for large scale recommendation tasks.

After the evaluation of GMF, we explored a reinforcement learning-inspired recommendation strategy using an **Epsilon-Greedy Bandit** framework. Rather than directly ranking all the papers globally, we implemented the recommendation as a two stage pipeline combining TF-IDF retrieval with adaptive reranking. First, TF-IDF cosine similarity was used to retrieve a candidate pool of semantically related papers for each query article. The epsilon-greedy algorithm then balanced exploration and exploitation by selecting high performing recommendations most of the time while also occasionally exploring less tested papers with probability ε. Reward estimates were updated incrementally based on whether recommended papers matched held out citation relationships. This method was particularly useful because research recommendation is dynamic, and highly relevant papers may initially appear less connected or less cited. By continuously exploring new candidate papers, the recommender system avoided repeatedly suggesting only highly popular works and improved recommendation diversity.

We then implemented **Neural Matrix Factorization (NeuMF)**, an advanced neural collaborative filtering architecture that combines both linear and nonlinear interaction modeling. NeuMF integrates two complementary components: a GMF branch that captures linear latent interactions and a Multi Layer Perceptron (MLP) branch that learns higher order nonlinear relationships between papers. Citation relationships were modeled as implicit feedback interactions, where embeddings passed through multiple neural layers with nonlinear activations before being combined into a final prediction layer estimating citation relevance scores. We further optimized training using Xavier weight initialization, mixed-precision GPU training, presampled negative examples, and cached embeddings for faster evaluation. By combining linear latent structure with deep nonlinear interaction learning, NeuMF provided a more expressive representation of paper relationships and improved recommendation accuracy beyond direct citation similarity.

After reaching satisfactory results**,** we pushed further into newer content of graph based models. To apply the PageRank algorithm, we had to set up the data into a DAG; we created edges by linking the paper_id directly with the cited_paper_id and the nodes being the unique arXiv ID. Using the networkx library, we were able to efficiently model the data and found the best model so far. We computed the importance scores by simulating a random walk over this graph, where importance flows through incoming citations and is reinforced by citations from already important papers. This is useful for our use case because it identifies globally influential or foundational papers that are widely referenced, independent of any specific query paper.

After further understanding of the PageRank algorithm, we noted that the standard PageRank algorithm would be biased towards more famous papers, while Personalized PageRank algorithm restarts at the query paper, allowing for the recommendations to be more accurate and even help with the cross-domain recommendations more. However, this algorithm is quite costly, so we implemented a variant of PPR using Monte Carlo, since this method gives us a good approximation via walk sampling. In this algorithm, we use the same citation graph as the standard PageRank, but simulate random walks starting from a specific query paper, repeatedly following citation links while occasionally restarting at the query node. We estimate paper relevance based on how frequently each node is visited during these walks. This approach works well because it captures local neighborhood structure in the citation network, effectively identifying papers that are closely connected to the query paper through citation paths, even if they are not directly linked. This method was a huge improvement from the previous standard PageRank Algorithm.

Additionally, upon evaluating the standalone TF-IDF and PageRank models, we developed a hybrid recommender that combined semantic similarity with graph based authority scoring. TF-IDF captured textual similarity between research papers using Bag of Words vectorization and cosine similarity, while PageRank identified globally influential papers within the citation network by modeling papers as nodes and citations as directed edges. We then combined these signals using a weighted scoring function, where the TF-IDF component emphasized semantic relevance and the PageRank component rewarded influential or highly connected papers. To prevent the recommendations from becoming overly biased toward highly cited papers, all scores were normalized and balanced carefully to preserve diversity and maintain stronger cross-domain recommendations. This hybrid model achieved the strongest overall performance across all tested systems, producing the highest values for Precision@10, Recall@10, NDCG@10, Hit Rate@10, and MRR@10, demonstrating that combining semantic relevance with citation authority significantly improved recommendation quality.

We also developed a TF-IDF + SVD hybrid recommender that combined latent collaborative filtering signals with semantic text similarity. To implement this model, we first constructed a sparse paper to paper citation matrix and applied truncated Singular Value Decomposition (SVD) to generate low dimensional latent embeddings for each paper. In parallel, TF-IDF vectors were computed from paper text and cosine similarity was used to measure semantic similarity between papers. The final recommendation score was calculated by using a weighted combination of SVD and TF-IDF scores, where the SVD component captured hidden citation patterns and latent research relationships while TF-IDF preserved semantic alignment between papers. Similar to the PageRank hybrid, score normalization was applied before combining the models to ensure balanced contributions from both collaborative filtering and content-based similarity signals.
Compared to the standalone SVD baseline, the TF-IDF + SVD hybrid produced substantial improvements across all evaluation metrics, including Precision@10, Recall@10, NDCG@10, Hit Rate@10, and MRR@10. However, the TF-IDF + PageRank hybrid consistently outperformed every other tested model, indicating that graph-based citation authority combined with semantic similarity was especially effective for research paper recommendation.

UI:

To make the recommendation system accessible without running the notebook, we built a self-contained Gradio web application (`app.py`) that loads pre-trained model artifacts from Google Drive and serves all nine trained models through a unified interface.
Tech Stack: Gradio 4.x - python native UI

Key UI Features

- **Single text input** for all models — users paste an abstract once and switch models freely without re-entering their query.
- **Domain filter checkboxes** — the eight arXiv domains (`cs`, `math`, `physics`, `stat`, `eess`, `q-bio`, `q-fin`, `econ`) can be selected to boost or filter results. Filtering uses prefix matching so selecting `cs` matches all `cs.*` subcategories.
- **Results table** — shows rank, arXiv ID, year, full category list, title (hyperlinked to `arxiv.org`), abstract snippet (300 chars), and model score.
- **Category distribution pie chart** — generated from the displayed recommendations, counting all categories of each result paper grouped by top-level domain.
- **Score formatting** — scores below 0.001 (e.g. PageRank values) are displayed in 6-decimal scientific format rather than rounding to zero.

IV. Benchmark & Evaluation

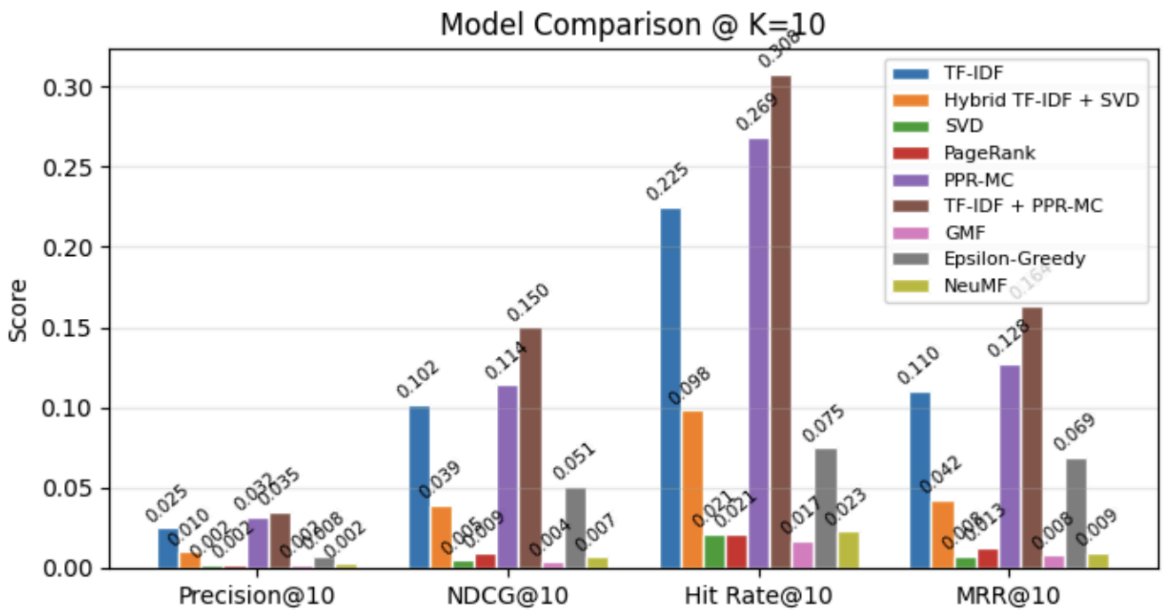
Evaluation:

★ Train/test split: 80% of citation edges for training, 20% held out for evaluation
★ Query papers: 1,000 randomly sampled papers with ≥1 held-out citation
★ Exclusions: self-links and training citations removed from ranked list
★ Top-K ranking: np.argpartition (O(n)) for speed, then sort only top-K
★ Metrics: Precision@K, Recall@K, NDCG@K, Hit Rate@K, MRR at K ∈ {5, 10, 20}

Model Performance Breakdown:

Below is a table comparing the metrics of all of the tested models, including the hybrid models

Model	**P@10**	**R@10**	**NDCG@10**	**HR@10**	**MRR@10**
SVD (baseline)	0.0021	0.0068	0.0048	0.021	0.0076
TF-IDF	0.0247	0.1486	0.1021	0.225	0.1103
NeuMF	0.0024	0.0093	0.0069	0.023	0.0094
PageRank	0.0021	0.0102	0.0089	0.021	0.0129
Monte PPR	0.0319	0.1632	0.1141	0.269	0.1277
GMF	0.0021	0.0042	0.0038	0.017	0.008
Epsilon-Greedy Bandit	0.0076	0.0488	0.0511	0.075	0.0692
TF-IDF+SVD	0.0103	0.0592	0.0393	0.098	0.0423
TF-IDF + PPR-MC	0.0351	0.2088	0.1501	0.308	0.1637



Graph 5

Analysis of Results:

- ★ Across all evaluated models, the **strongest** performance was achieved by the **hybrid TF-IDF + Monte Carlo PPR model,** which obtained the highest Precision@10 (0.0351), Recall@10 (0.2088), NDCG@10 (0.1501), and Hit Rate@10 (0.3080).
- ★ Monte Carlo Personalized PageRank also performed strongly on its own, demonstrating that citation graph structure is a powerful signal for identifying relevant research papers.
- ★ The TF-IDF baseline remained highly competitive, outperforming most collaborative filtering and neural approaches, highlighting the strength of semantic similarity in academic text
- ★ In contrast, matrix factorization and neural models (SVD, GMF, NeuMF) showed weak performance, primarily due to extreme sparsity in the citation interaction matrix and limited signal for learning stable embeddings.
- ★ PageRank-based methods performed modestly, indicating that global citation importance alone is insufficient for personalized recommendation.
- ★ The Epsilon-greedy bandit model achieved moderate results, suggesting the model is more suitable for exploration and diversity rather than precise citation prediction.
- ★ Overall, results indicate that hybrid models combining content-based and graph-based signals consistently outperform individual methods, reinforcing the importance of multi-signal.

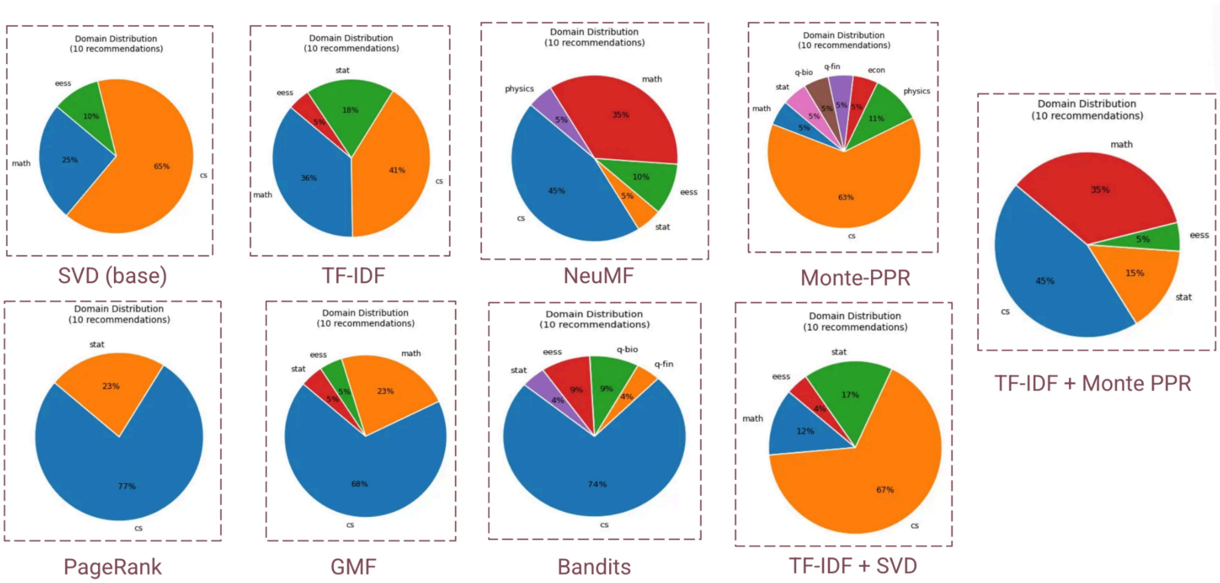
Applying the models in real-time:

Given this prompt,

- ★ abstract: This paper investigates machine learning approaches for solving partial differential equations (PDEs) commonly found in physical and engineering systems. We evaluate neural network architectures that incorporate mathematical constraints directly into training and compare performance against traditional numerical optimization methods. The objective is to improve computational efficiency while maintaining solution accuracy and generalization.

- Math + CS prompt
- No category boost added

The recommendation's category distribution per model is as follows:



Analysis:

The domains represented include Computer Science (cs), Mathematics (math), Statistics (stat), Economics and Social Sciences (eess), Physics, q-fin, and q-bio. These visualizations provide insight into how different recommendation techniques balance relevance, diversity, and specialization across academic fields.

The main observation across nearly all models is the dominance of the Computer Science domain. Most recommendation approaches allocate a significant proportion of recommendations to CS related content, indicating either dataset imbalance or an inherent popularity bias within the algorithms. However, the extent of this dominance varies considerably depending on the recommendation strategy used.

The **SVD (Singular Value Decomposition)** baseline demonstrates a strong concentration toward Computer Science, which accounts for 65% of all recommendations. Mathematics contributes 25%, while eess receives only 10%. This distribution suggests that SVD primarily reinforces dominant interaction patterns present in the training data. Since SVD is a collaborative filtering method, it relies heavily on historical user-item interactions, making it susceptible to popularity bias. Consequently, niche or underrepresented domains receive limited exposure.

In contrast, the **TF-IDF** model produces a significantly more balanced recommendation profile. Computer Science decreases to 41%, while Mathematics rises to 36% and Statistics to 18%. The inclusion of EEES, though small, further improves diversity. TF-IDF is a content-based approach that evaluates textual similarity between documents, enabling it to recommend semantically related content rather than relying solely on popularity. This results in broader interdisciplinary coverage and demonstrates the advantage of content-aware methods in improving recommendation diversity.

The **PageRank** algorithm exhibits the highest level of concentration among the first group of models. Approximately 77% of recommendations belong to Computer Science, while the remaining 23% are from Statistics. No other domains appear. Since PageRank emphasizes graph centrality and connectivity, it tends to amplify highly connected nodes within the recommendation graph. As a result, dominant research communities become overrepresented, reducing exploration of less connected domains.

Similarly, **GMF (Generalized Matrix Factorization)** strongly favors Computer Science, which represents 68% of recommendations. Mathematics contributes 23%, while Statistics and eess each account for only 5%. Although GMF extends traditional matrix factorization through neural modeling, it still prioritizes strong interaction signals, leading to a concentration similar to other collaborative filtering approaches.

The second figure introduces additional recommendation methods and hybrid approaches. Among these, **NeuMF (Neural Matrix Factorization)** demonstrates one of the most balanced domain distributions. Computer Science accounts for 45%, Mathematics 35%, eess 10%, and both Statistics and Physics 5% each. Compared to GMF, NeuMF achieves better diversity because its neural architecture captures more complex nonlinear relationships between users and items. This enables broader generalization and reduces excessive specialization.

The **Monte-PPR (Monte Carlo Personalized PageRank)** model remains strongly centered on Computer Science, which occupies 63% of recommendations. Physics contributes 11%, while smaller domains such as economics, q-fin, and q-bio each appear at approximately 5%. Although Monte-PPR still favors highly connected domains, it introduces a modest degree of exploration by exposing users to niche subjects. This suggests that random-walk-based methods can improve diversity slightly while maintaining graph relevance.

A particularly effective approach is the **TF-IDF + Monte PPR hybrid model**. This method balances recommendations across multiple domains: Computer Science represents 45%, Mathematics 35%, Statistics 15%, and eess 5%. By combining semantic similarity from TF-IDF with graph propagation from Monte-PPR, the model achieves both relevance and diversity. The hybrid structure mitigates the popularity bias seen in pure graph-based methods while preserving meaningful relational information.

The **Bandits** algorithm displays another highly concentrated distribution, with 74% of recommendations belonging to Computer Science. Smaller contributions come from eess, q-bio, q-fin, and Statistics. Bandit-based approaches optimize recommendations by exploiting historically successful choices. While effective for maximizing immediate reward or engagement, this strategy often limits exploration and reinforces already dominant domains.

Finally, the **TF-IDF + SVD** hybrid model shows moderate improvement over standard SVD. Computer Science remains dominant at 67%, while Statistics contributes 17%, Mathematics 12%, and eess 4%. The inclusion of TF-IDF introduces some semantic diversity, but the collaborative filtering component still heavily influences the final recommendations.

Overall, the figures highlight important trends in recommendation system behavior. Pure collaborative filtering and graph-centrality methods tend to amplify dominant fields such as Computer Science, often at the expense of interdisciplinary diversity. In contrast, content-based approaches like TF-IDF distribute recommendations more evenly across domains. Hybrid systems that combine semantic understanding with graph or latent-factor methods achieve the best balance between relevance and diversity.

Among all models, **TF-IDF**, **NeuMF**, and **TF-IDF + Monte PPR** emerge as the most balanced recommendation strategies. These approaches successfully reduce over-specialization and encourage broader academic exploration. On the other hand, methods such as **PageRank**, **Bandits**, and **GMF** demonstrate strong popularity bias and limited diversity. Therefore, for recommendation systems intended to support interdisciplinary discovery and balanced exposure, hybrid and semantic-aware methods appear to be the most effective solutions.

All prompts written for demo:

- (econ + cs) prompt 1:
    - title: Multi-Modal Deep Learning for Stock Market Forecasting Under Economic Uncertainty
    - abstract: This paper proposes a framework for forecasting stock market movements by combining historical price data, macroeconomic indicators, and financial news sentiment. We evaluate transformer and recurrent architectures for predicting market volatility and directional movement under changing economic conditions. The objective is to improve robustness across market regimes while maintaining interpretability.
- (phys + cs) prompt 2:
    - title: Deep Learning Methods for Particle Event Classification in High-Energy Physics
    - abstract: This work investigates deep neural architectures for identifying and classifying particle collision events. The study evaluates representation learning and uncertainty estimation methods to improve event detection accuracy under noisy experimental conditions.
- (bio + cs) prompt 3:
    - title: Graph-Based Representation Learning for Predicting Protein Interaction Networks
    - abstract: This paper proposes a graph representation framework for identifying functional relationships between proteins. Network structure and biological metadata are incorporated to improve prediction of previously unknown interactions.

- ★ (cs only) prompt 4:
    - title: Improving Vision Transformer Performance for Fine-Grained Image Classification
    - abstract: This paper investigates methods for improving fine-grained image classification using Vision Transformer architectures. We evaluate attention mechanisms, self-supervised pre-training, and optimization strategies to improve generalization under limited labeled data settings.
- ★ (math + cs) prompt 5 (used above)
    - title: Neural Approaches for Solving Partial Differential Equations Using Physics-Informed Learning
    - abstract: This paper investigates machine learning approaches for solving partial differential equations (PDEs) commonly found in physical and engineering systems. We evaluate neural network architectures that incorporate mathematical constraints directly into training and compare performance against traditional numerical optimization methods. The objective is to improve computational efficiency while maintaining solution accuracy and generalization.

V. Conclusions

Challenges:

- ★ Category matching bug
    - The arXiv OAI-PMH API offers two metadata formats. The default format (oai_dc) stores categories as human-readable strings like "Computer Science - Machine Learning", not as codes like cs.LG.
    - Our regex filter matched only 1-2 records per set as a result. Switching to metadataPrefix="arXiv" fixed this because that format provides a dedicated <categories> field with space-separated arXiv codes that match exactly.
- ★ Zero in-sample citations after fetching 3.2 million raw links
    - This was the most confusing bug. The root cause was two-fold. First, pandas silently reads arXiv IDs like "2301.12345" as float64 from CSV files because they look like numbers.
    - A string comparison between a float and a string always returns False, so isin() reported no matches. Fixing this required adding dtype=str to every CSV read of ID columns.
    - Second, if the Semantic Scholar checkpoint was saved during a previous run with a different set of papers, the source IDs in the checkpoint would not match the current paper set. We added a staleness check that discards the checkpoint if fewer than 50% of its IDs overlap with the current run.
- ★ Evaluation speed
    - The initial TF-IDF evaluation loop processed one query paper at a time, calling toarray() on a 163k-element sparse vector and then running np.argsort on all 163k scores. At 1 second per paper that was 17 minutes for 1,000 queries.
    - Batching 100 queries into a single sparse matrix multiply and replacing argsort with argpartition (which only finds the top-K elements in O(n) instead of sorting everything in O(n log n)) brought the runtime under 2 minutes.
- ★ Session crashes during long fetches

- The Semantic Scholar fetch takes several hours with a free API key. Colab sessions disconnect mid-run regularly. We solved this by writing checkpoints atomically: every N batch, the code writes to a .tmp file and then renames it into place.
- A rename is atomic on all major filesystems, so a crash mid-write never leaves a corrupt checkpoint that would be silently loaded on the next run. The arXiv fetch uses the same pattern per (set, year) task.

★ Cold-start papers
- About 18.5% of papers are never cited by any other paper in the dataset. Keeping them in the training graph adds noise without adding signals for the CF models.
- The cold-start filter drops papers whose total degree (in-degree plus out-degree) is below 2, which removes roughly 7k papers while keeping the 163k that have at least some citation connectivity.

Takeaways:

★ Different Algorithms
- TF-IDF captured semantic similarity directly from article abstracts through content-based filtering
- SVD, GMF, and NeuMF learned latent relationships between papers using citation interactions through collaborative filtering approaches
- PageRank & Personalized PageRank leveraged citation graph structure to identify influential and query-specific relevant papers
- Bandit methods introduced an exploration-exploitation strategy
- Combining these recommendation signals enabled richer recommendations and improved cross-domain discovery, as demonstrated in the project use cases

★ Future Work
- Extend the hybrid recommender by incorporating stronger model combinations and additional neural recommendation approaches
- Include the full arXiv dataset, including the papers we removed via cold start filter
- Improve the user interface through additional recommendation modes, including category-only recommendations and user being able to combine models
- Explore methods for reducing cold-start limitations while maintaining recommendation quality at scale

★ Data Preparation & Scalability
- Recommendation quality depended heavily on accurate metadata extraction and reliable citation alignment
- Data validation, checkpointing, and recovery mechanisms were necessary to support large-scale data collection and preprocessing
- Efficient preprocessing pipelines and sparse representations were required to operate on large academic datasets
- Scalable evaluation methods were necessary to support experimentation across multiple recommendation algorithms

★ Hybrid Recommender vs Individual Algorithms
  ○ Combining multiple recommendation approaches produced stronger recommendations than relying on a single model by far.
  ○ For practical deployment, efficient approaches such as Monte Carlo Personalized PageRank combined with lightweight recommendation models are likely the most suitable solution due to lower computational cost and faster response times
  ○ Hybrid recommendation enabled both semantic relevance and citation-aware discovery, supporting cross-domain research exploration.

VI. Source Code

★ GitHub repository: https://github.com/Talina06/research-paper-recommender-system
★ Slides: Cmpe 256 slides Arxivists

File	Contents
notebook.ipynb	Full pipeline: data collection, EDA, baseline model, evaluation framework
app.py	Gradio web UI
README.md	Setup and run instructions

VII. Contribution Table

Name	**Contributions**
Manjula	Models: GMF, Epsilon-Greedy, NeuMF, TF-IDF-SVD Hybrid model Report: Model breakdown, Distribution model analysis Partial EDA
Sheetal	★ Project Idea, Proposal & Prototypes on setting up the data (citation x citation matrix) ★ Models: SVD, PageRank, MonteCarlo PageRank ★ PPT Slides ★ Model testing in UI + Prompts for demo ★ Majority of Report
Talina	• Data Gathering, Merge and Preparation • EDA • Evaluation Methods • Baseline Model TF-IDF • Hybrid Model (SVD + PPR-MC) • UI